

01_TECHNICAL_COMPLIANCE_BRIEF: Aura-Guard v1.3

Standard: Atom-Grade Deterministic Trust Infrastructure

Classification: Confidential

Date: 12 maja 2026

1. Introduction

Aura-Guard v1.3 is a deterministic AI audit middleware designed to provide an immutable evidence chain for organizations using generative AI. It addresses the "Black Box" problem by ensuring that every interaction is governed by signed policies and recorded in a cryptographically secured audit lineage.

2. Core Architectural Pillars

2.1 Deterministic Decision Engine

The engine is built in Rust to ensure memory safety and high performance. Unlike probabilistic filters, Aura-Guard guarantees that: **Input (A) + Policy (P) = Decision (D) + Hash (H)**. This result is identical every time it is executed, enabling perfect replayability.

2.2 Cryptographic Hash Chaining (The Lineage)

Every audit record in version 1.3 is part of a secure chain. Each entry contains a `chain_hash`, which is computed from the current decision and the hash of the *previous* record. This creates a tamper-evident sequence where deleting or altering a single entry invalidates the entire audit history.

2.3 Policy Integrity (Ed25519 Signatures)

Policies are no longer just YAML files. In v1.3, every policy must be accompanied by a digital signature (Ed25519). The system verifies the signature against a trusted public key before loading. This prevents "Unauthorized Policy Injection" or manual tampering by system administrators.

3. Technical Security Features

3.1 Unicode Normalization & Hardening

To prevent bypass attacks (e.g., using hidden zero-width characters or visually similar homographs), Aura-Guard performs:

- **NFC Normalization:** Standardizing Unicode characters.
- **Hidden Character Stripping:** Removing characters like `\u{200B}` (Zero Width Space) before evaluation.
- **Case Normalization:** Ensuring rules trigger regardless of capitalization while preserving original text for the hash record.

3.2 Fail-Closed Execution Posture

If Aura-Guard detects an integrity violation (e.g., an invalid policy signature, a broken hash chain in the log, or a malformed regex), it enters a **Fail-Closed** state. It halts the audit process and returns a critical error, ensuring that no unverified or compromised AI traffic is permitted.

4. Regulatory Compliance Mapping

Regulation	Requirement	Aura-Guard Implementation
EU AI Act (Art. 12)	Automatic recording of events (Logs).	Immutable, cryptographically chained audit logs.
EU AI Act (Art. 13)	Transparency and information to users.	Full reproducibility of decisions through Replay CLI.
DORA	Operational resilience and auditability.	Air-gapped deployment with local-first evidence storage.
GDPR	Data Minimization & Privacy.	Stateless processing; no PII leaves the local infrastructure.

5. Deployment & Integration

Aura-Guard v1.3 is a self-contained binary. It does not require internet access, external databases, or GPUs. It can be deployed in highly secured environments (Air-Gapped) and integrated into existing AI pipelines via a simple mTLS-secured REST API.

6. Replay Verification

The `aura-replay` tool allows external auditors to verify the integrity of the audit log without needing access to the live system. By validating the hash chain and comparing it against the original evidence, auditors can prove that the organization has remained in compliance over any given period.

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